Curriculum vitae

Suresh Perumal, Ph.D.,
Assistant Professor,
Department of Materials Science and Metallurgical Engineering,
Indian Institute of Technology (IIT), Hyderabad.
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Research Interests:

- **Thermoelectric Materials and Devices:** *Silicides / Chalcogenides / Antimonides* / Oxides
- **Magnetic Refrigeration:** *NiMnIn-based Heusler Alloys*
- **TE Metrology:** Design and Fabrication of Transport Property Measurement Systems
- **Magneto-thermoelectric Materials.**

Education:

- **Ph.D.,** Materials Research Centre, **Indian Institute of Science (IISc),** Bangalore (**2013**)
  (Title: *Nanostructurization of Metal Silicides for High-Temperature Thermoelectric Materials*)
- **M.Sc.,** Physics, **University of Madras,** Chennai, Tamilnadu (**2007**) – **[Outstanding*]**
- **B.Sc.,** Physics, **Periyar University,** Salem, Tamilnadu (**2005**)- **[Distinction*]**

Professional experience:

- **Assistant Professor, Materials Science & Metallurgical Engineering, IIT Hyd.** (April,2023 –**Present**).
- **Assistant Professor, Metallurgical and Materials Engineering, IIT, Jodhpur** (Jan.2023 –April 2023).
- **Assistant Professor, SRM IST,** Kattankulathur, Chengalpattu, Tamilnadu (June, 2018 –Jan, 2023).
- **Postdoctoral fellow, JNCASR,** Jakkur, Bangalore-560 064, India. (June, 2016 – June, 2018).
- **Postdoctoral fellow, SKKU,** Suwon, South Korea. (Jan., 2016 – May, 2016).
- **Postdoctoral fellow, JNCASR,** Jakkur, Bangalore-560 064, India. (Dec., 2014 – Jan, 2016).
- **Postdoctoral fellow, Laboratory CRISMAT,** ENSI-CAEN, Caen, France (Oct., 2013 – Oct., 2014).
- **Research Associate (Junior & Senior), MRC, IISc,** Bangalore-560 012, (July, 2012 – Oct., 2013).

Professional membership:

- Life membership (**#59468**) in the Indian Institute of Metals **(IIM)**
- Life membership ***(LMB- 3377)*** in Materials Research Society of India **(MRSI)**
- Life membership ***(SL- 454)*** in Indian Ceramic Society **(InCerS)**
- Life membership ***(LM 2011)*** in Electron Microscope Society of India **(EMSI)**

Ph.D. students (Ongoing and Guided)

- **IITH : Ongoing: 04 (PhD)**
- **SRM IST: Completed: 03**
  [Dr. V. Shaleni (2023), **Dr. M. Manoikumar** (2023), and **Dr. G. S. Madhuvathani** (2023)**]**

Awards/Recognition:

- *Top 2% Scientists listed from Stanford University-**2025***
- *Editorial board member – **Scientific Report** [IF: 3.8] – Nature Publication-**2024***
- *Top 2% Most cited Scientists, Standford University List- **2024***
- *Youth Editorial Board Member in **Journal of Materiomics [IF. 8.3]-2023***
- *Top 2% most-cited list from Stanford University--**2023***
- *Institute Junior and Senior **Research Associate fellowship** from Aug., 2012- Oct., 2013.*
- ***Prof. Selvarajan award** for International Scientific Travel- 2011.*
- ***Dr. S. Kalyanaraman** Memorial endowment Prize-Certificate of merit in M. Sc., 2007.*

- ### Sponsored Research Project
  - **Principal Investigator (PI):** ***Project title***:*"*Microstructural Engineering in Higher Manganese Silicide: Novel approaches for Eco-friendly and High-performance Thermoelectric Power Generation **(MEET)**. **Agency: CEFIPRA Duration: 2024-2027 Grant: 160.00 Lakhs [Ongoing]**
  - **Principal Investigator (PI):** ***Project title***: "Development of Nanostructured Higher Manganese silicide-based Thermoelectric Power Generators to harvest the waste heat from car exhausters. (**SG-167**). **Agency: IIT-SEED Duration: 2023-2026 Grant: 30.00 Lakhs [Ongoing]**
  - **Principal Investigator (PI):** ***Project title***: "Fabrication of Nano-structured $MnSi_{\delta}$/Si based Segmented Thermoelectric Devices: Towards Low–cost and Eco-Friendly Power Generators by recovering waste heat from car exhausters and power plants". **Agency: DST-SERB Duration: 2019-2022 Grant: 49.46 Lakhs [Completed]**
- ### Book chapter
  - Ananya Banik, **Suresh Perumal** and Kanishka Biswas, *Thermoelectric properties of Selected Metal Chalcogenides nanosheets/nanofilms grown by Chemicals and Physical routes, pp. **157-184**,* **by Springer Nature Publication.** (**Book title**:*"Thermoelectric thin films: Materials and Devices")*
  - G. S. Madhuvathani, M. Manojkumar, Arthiya S E, **Suresh Perumal*, Thermoelectric Materials for Energy Conversion, Submitted to by Springer Nature Publication. [Under revision] – 2024.** (**Book title**:*" Design and Advanced Materials for Energy Applications")*

Peer-reviewed publications

- ***Total publications: 79*** *Citations: **3308*** *H- Index: **27*** *i-10- Index: **43***
  (Source: Google Scholar citations)

After joining IITH:

1. Akshara Dadhich, Kaushalya Kumari, Achintya Kumar Patra, Bhuvanesh Srinivasan, **Suresh Perumal,** MS Ramachandra Rao, Kanikrishnan Sethupathi, Realizing high thermoelectric performance in MXene-incorporated $Yb_{0.4}Co_{3.96}Ti_{0.04}Sb_{12}$ via carrier and phonon engineering, ***Journal of Materials Chemistry A,*** 13, 16770-16784, **2025.** [IF: **9.5** and **Q1** Journal]
2. A Dadhich, B Srinivasan, **Suresh Perumal,** MSR Rao, K Sethupathi, Dual doping strategy for enhancing the thermoelectric performance of $Yb_{0.4}Co_4Sb_{12}$, ***Journal of Materials Chemistry C,*** 13 (14), pp. 7368-7379, **2025.** [IF: **9.5** and **Q1** Journal]
3. Debattam Sarkar, Subarna Das, Vaishali Taneja, Manisha Samanta, Koushik Jagadish, Animesh Das, Monika Bhakar, Prasad VD Matukumilli, **Suresh Perumal,** Goutam Sheet, Dirtha Sanyal, Koushik Pal, N Ravishankar, Umesh V Waghmare, Kanishka Biswas, Glassy Thermal Transport Triggers Ultra-High Thermoelectric Performance in GeTe, ***Advanced Materials,*** 37 (12), 2417561, **2025**. [IF: **26.8** and **Q1** Journal]
4. Premanand Ganesan, Sridevi D V, Manojkumar Moorthy, **Suresh Perumal,** Silambarasan S, Thandavarayan Maiyalagan, Tushar H. Rana, Arun Prakash Periasamy, Ramesh V, High-Powered Nanostructured rGO/g-C_3N_4/CoSe||AC Electrodes Employed in an Asymmetric Supercapacitor Device, ***Energy Storage,*** 7 (6), e70257, **2025.** [IF: **4.1** and **Q1** Journal]
5. A. Kumararaj, **Suresh Perumal,** K. B. Karuppanan, G. Arunachalam, Electrochemical Performance of TiO_2/Mn_2O_3 Nanocomposite in Different Electrolyte Environments, ***ChemNanoMat,*** 11, e202400207, **2025**. [IF: **2.6** and **Q2** Journal]
6. T Ramesh, **Suresh Perumal,** KK Bharathi, G Sankar, G Arunachalam, Structural and electrochemical characterization of Zn-doped $Ca_3Co_4O_9$ for supercapacitor applications, ***Journal of Materials Science***, pp. 1-21, **2025**. [IF: **3.9** and **Q1** Journal]
7. L Ravi, PW Vanaraj, BS Subathra, **Suresh Perumal,** S Kumar, Enhancing mechanical, and tribological properties of aluminum metal matrix composite reinforced with high entropy alloy using friction stir processing, ***Materials Chemistry and Physics***, 338, 130614, **2025**. [IF: **4.7** and **Q1** Journal]
8. Arunaprabha Andipet, Prince Wesley Vanaraj, Lokeswaran Ravi, Kamala Bharathi Karuppanan, Krishnamoorthy Guruvidyathri, **Suresh Perumal*,** Ravi Kirana, High Electrochemical Performance and Photocatalytic Activity in Multicomponent Nanostructured (CuZnNiCoMg)O, ***Inorganic Chemistry,*** 64 (25) 12492–12506, **2025.** [IF: **4.7** and **Q1** Journal]
9. Prince Wesley Vanaraj, Lokeswaran Ravi, Shashikant P Patole, Dalaver H Anjum, SN Megha, Mangalampalli SRN Kiran, **Suresh Perumal,** Ravikirana, Impact of atomic size misfit on lattice distortion in AlFeCoNiX (X=

Cr/Mn/Zr) multicomponent alloys, ***Journal of Materials Science,*** *60*, 4367–4388, **2025.** [IF: **3.9** and **Q1** Journal]

10. Tamilarasi Ramesh, **Suresh Perumal,** Kamala Bharathi Karuppanan, Geetha Arunachalam, Realizing high electrochemical performance in layered polycrystalline $Ca_3Co_4O_9$ oxide, ***Emergent Materials,*** 8, 289–300, **2025.** [IF: **4.1** and **Q2** Journal]
11. Jothilal Palraj, Muhammad Sajjad, Manojkumar Moorthy, Madhuvathani Saminathan, Bhuvanesh Srinivasan, Nirpendra Singh, Rajasekar Parasuraman, Shashikant P Patole, Kiran Mangalampalli, **Suresh Perumal*,** High thermoelectric performance in p-type ZnSb upon increasing Zn vacancies: an experimental and theoretical study, ***Journal of Materials Chemistry A*, **12, 13860-13875, **2024**. [IF: **10.7** and **Q1** Journal]
12. John D Rodney, Sindhur Joshi, Subhasmita Ray, Lavanya Rao, S Deepapriya, Karel Carva, Badekai Ramachandra Bhat, NK Udayashankar, **Suresh Perumal**, Sadhana Katlakunta, C Justin Raj, Byung Chul Kim, Electrocatalytic synergies of melt-quenched Ni-Sn-Se-Te nanoalloy for direct seawater electrolysis, ***Chemical Engineering Journal*,** 499, 155775, **2024.** [IF: **13.4** and **Q1** Journal]
13. Madhuvathani Saminathan, Saravanan Muthiah, Rajasekar Parasuraman, Debattam Sarkar, Kiran Mangalampalli, **Suresh Perumal*,** Realizing low thermal conductivity in Cr-doped nanostructured higher manganese silicide, ***Ceramics International,*** 50, 33599-33606, **2024.** [IF: **5.1** and **Q1** Journal]
14. A Abhishek, Tushar H Rana, Rajasekar Parasuraman, **Suresh Perumal**, V Ramesh, Thermoelectric, mechanical and electrochemical properties of pure single-phase FeSb, ***Ceramics International,*** 50, 26760-26769, **2024.** [IF: **5.1** and **Q1** Journal]
15. Gouri Sankar, Madhuvathani Saminathan, **Suresh Perumal,** R Tamilarasi, Geetha Arunachalam, Enhanced thermoelectric properties of Cu 1.8 S via the introduction of ZnS nanostructures, ***Sustainable Energy & Fuels*,** 8 (23), 5514-5523, **2024.** [IF: **5.0** and **Q1** Journal]
16. Moorthy, Manojkumar; Govindaraj, Prakash; Parasuraman, Rajasekar; Bhui, Animesh; Gadhavajhala, Sri Sai Samhitha; Srinivasan, Bhuvanesh; Venugopal, Kathirvel; **Suresh Perumal*,** Sulfur vacancies driven band splitting and phonon anharmonicity enhance the thermoelectric performance in n-type $CuFeS_2$, ***ACS Appl. Energy Mater.*,** 7, 2008–2020, **2024**. [IF: **5.1** and **Q1** Journal]
17. Gouri Sankar, Madhuvathani Saminathan, **Suresh Perumal,** Geetha Arunachalam, Thermoelectric properties of aliovalent Zn doped $Cu_{1.8}S$ polycrystalline materials, ***Ceramics International,*** 50, 13400-13411, **2024**. [IF: **5.1** and **Q1** Journal]
18. B. S. Subathra, Madhuvathani Saminathan, Prince Wesley, Lokeshwaran Ravi, Manjusha Battabyal, Debattam Sarkar, **Suresh Perumal*,** Ravikirana*, Thermoelectric properties of iso-valent Bi substituted n-type Ti2NiCoSnSb high entropy alloys, ***Intermetallics,*** 167, 108233, **2024**. [IF: 4**.4** and **Q1** Journal]
19. Aradhana Acharya, Barnasree Chanda, Madhuvathani Saminathan, **Suresh Perumal**, K Jayanthi, K Annapurna, NM Anoop Krishnan, Bhasker Gahtori, Milan Kanti Naskar, Srabanti Ghosh, Amarnath R Allu, Suman Kumari Mishra, *Influence of metal organic framework glasses on thermoelectric properties of* $AgSb_{0.96}Zn_{0.04}Te_2$ *alloy*, ***J. Non-Cryst. Solids*,** 627, 122816, **2024**. [IF: **3.2** and **Q2** Journal]
20. A Abhishek, V Ramesh, **Suresh Perumal,** Prasant Kumar Nayak, Structural, vibrational and electrochemical studies of bulk and nano SnSb for supercapacitor application, ***J. Alloys. Compds.*,** 969, 172293, **2023.** [IF: **5.8** and **Q1** Journal]
21. Akshara Dadhich, Madhuvathani Saminathan, Saravanan Muthiah, Animesh Bhui, **Suresh Perumal,** MS Ramachandra Rao, Kanikrishnan Sethupathi, *Enhancement in Thermoelectric Performance in Ti-doped* $Yb_{0.4}Co_4Sb_{12}$ *Skutterudites via Carrier Optimization and Phonon Anharmonicity*, **ACS Appl. Mater. Interfaces,** 15 (45), 52368-52380, **2023.** [IF: **8.5** and **Q1** Journal]
22. Akshara Dadhich, Madhuvathani Saminathan, Kaushalya Kumari, **Suresh Perumal*,** MS Ramachandra Rao*, K Sethupathi*, *Physics and Technology of Thermoelectric Materials and Devices*, ***J. Phys. D: Appl. Phys.,*** 56, 333001, ***2023***. [IF: 3.1 and **Q2** Journal]
23. Hajeesh Kumar Vikraman, Jeena George, Rence P Reji, Guru Prasad Kuppuswamy, Sanjay D Sutar, Anita Swami, Sharmiladevi Ramamoorthy, Anandhakumar Sundaramurthy, Sumit Pramanik, Surya Velappa Jayaraman, **Suresh Perumal**, Yuvaraj Sivalingam, SRN Kiran Mangalampalli, *Unprecedented Multifunctionality in Novel Monophase Micro / Nanostructured Ti-Zn Alloy*, ***Small,*** 2305126, **2023.**

Before joining IITH:

24. V Ravisankar, V Ramesh, B Gunasekaran, **Suresh Perumal,** P Karuppasasmy, T Kamalesh, *Centrosymmetric structure of novel barium (II)-dibenzo-15-crown-5-ether-zinc (II)-tetra-thiocyanate single crystal for nonlinear optical application*, *Opt. Quantum Electron.,* 55 (11), 953, **2023**.

25. T. Kayalvizhi, Ayyappan Sathya, **Suresh Perumal,** KRS Preethi Meher, Structural, Optoelectronic and Electrochemical behavior of the mechanochemically synthesized $CsPb_{1-x}Na_xBr_3$ (x = 0 to 0.15), **J. Sol. State Chem.**, 323, 123997, **2023.**
26. Manojkumar Moorthy, Bhuvanesh Srinivasan, David Berthebaud, Rajasekar Parasuraman, Suresh Perumal*, Enhanced Thermoelectric Performance and Mechanical Property in Layered Chalcostibite $CuSb_{1-x}Pb_xSe_2$, ***ACS Appl. Energy Mater.*** 6, 2, 723-730, **2023.**
27. Shaleni Venkatesan, E Meher Abhinav, N Pavan Kumar, S Kavita, M Manivel Raja, Rajasekar Parasuraman, Durai Murugan Kandhasamy, Suresh Perumal*, Realization of Enhanced Refrigeration Capacity with Non-Hysteretic Behavior in $Ni_{40}Gd_5Co_5Mn_{37}In_{12}Si_1$ Alloy Near Room Temperature at 2T, ***ACS Appl. Energy Mater.*** **5**, 12, 15959–15971, **2022.**
28. P. Ilayaraja, V. Ramesh, Suresh Perumal, D. V. Sridevi, Junaid Masud Laskar, K. Rajarajan, G. Anbalagan, Growth, optical, mechanical, dielectric properties of UV-nonlinear optical single-crystal of $[(NH_4)(Cd(NCS)_3)]\cdot C_{12}H_{24}O_6$, ***Mater. Sci. Eng., B.,*** 286, 116053, **2022.**
29. P Umadevi, KT Ramya Devi, DV Sridevi, Suresh Perumal, V Ramesh, Structural, morphological, optical, photocatalytic activity and bacterial growth inhibition of Nd-doped TiO_2 nanoparticles, ***Mater. Sci. Eng., B.,*** 286, 116018, **2022.**
30. G Premanand, DV Sridevi, **Suresh Perumal,** T Maiyalagan, John D Rodney, V Ramesh, New hybrid semiconducting CdSe and Fe doped CdSe quantum dots based electrochemical capacitors, ***Mater. Sci. Eng., B.,*** 286, 116015, **2022.**
31. S. R. N. Kiran Mangalampalli, Pamu Dobbidi Lakshmi, Narayan Ramasubramanian, Eswara Prasad Korimilli, **Suresh Perumal,** Srinivasa RaoBakshi, Advances in functional and structural ceramics: Development, characterization, and applications, **Ceram. Int.,** 48, 28763-28765, **2022.**
32. Manojkumar Moorthy, Animesh Bhi, Manjusha Battabyal, Suresh Perumal*, Nanostructured $CuFeSe_2$ Eskebornite: An efficient thermoelectric material with ultra-low thermal conductivity, ***Mater. Sci. Eng., B.,*** 248,115914, **2022.**
33. Madhuvathani Saminathan, Saravanan Muthaiah, Lokeswaran Ravi, Animesh Bhui, Reeshma Rameshan, Ravikirana, and Suresh Perumal*, Improved Thermoelectric properties of Fe-doped Si-rich Higher Manganese Silicides, **Mater. Sci. Eng., B.,** 284, 115912, **2022.**
34. John. D. Rodney, S. Deepapriya, S. Jerome Das M. Cyril Robinson, Suresh Perumal, Sadhana Katlakunta, Periyasamy sivakumar, Hyun Jung, and C. Justin Raj, Boosting overall electrochemical water splitting via rare earth doped cupric oxide nanoparticles obtained by co-precipitation technique, ***J. Alloys. Compds.,*** 921, 165948, **2022**.
35. DV Sridevi, Ramesh Vadivel, Suresh Perumal, E Sundaravadivel, V Sivaramakrishnan, A facile synthesis of Mn-doped ZnSe nanoparticles for enhanced photocatalytic activity and biological applications, ***Ceram. Int.,*** 48,29394-29402, **2022**.
36. Jothilal Palraj, Manojkumar Moorthy, Sadhana Katlakunta, **Suresh Perumal*,** Isovalent Bi substitution induced low thermal conductivity and high thermoelectric performance in *n*-type InSb, ***Ceram. Int.,*** 48, 29284-29290, **2022**.
37. Shaleni V, ES Kavita, Suresh Perumal*, Structure stability driven large magnetocaloric response in Ni–Co–Mn–In–Si Heusler alloy, ***Ceram. Int.,*** 48, 29059-29066, **2022**.
38. Manojkumar Moorthy, Jothilal Palraj, Lokesh Kannan, Sadhana Katlakunta, **Suresh Perumal*,** Structural, microstructural, magnetic, and thermoelectric properties of bulk and nanostructured *n*-type $CuFeS_2$ Chalcopyrite, ***Ceram. Int.,*** 48, 29039-29048, **2022**.
39. Lourdu Jame, Suresh Perumal, Manojkumar Moorthy, Joselin Retna Kumar, Nanocomposites of GO/D-Mannitol Assisted Thermoelectric Power Generator for Transient Waste Heat Recovery, ***J. Nanomater.*** 2022, 1-9, ***2022.***
40. Deepapriya, John D Rodney, S. Jerome Das, S. Lakshmi Devi, P. Nagaraj, J. R. Anusha, Suresh Perumal, J. Ermine Jose, C. Justin Raj, Synergetic effects of lanthanum substituted Ni-Zn-Cu-Co ferrite nano-composite with enhanced NH_3 sensing performance, ***J. Environ. Chem. Engineering,*** 9, 106829, **2021**.
41. Shaleni Venkatesan, E. Abhinav meher, S. Kavita, N. Pavankumar, M. Manivelraja and Suresh Perumal*, Enhanced Refrigeration Capacity of Ni-Mn-In-Si half-Heusler alloys for Magnetic refgrigerants, ***ECS J. Sol. State Sci. Techol.*** 10, 091009, **2021**.
42. J. Devakumar, V. Ramesh, Suresh Perumal, P. Anto Mariya Jeraldine, P. Amaladass, and R. Jaya Santhi, Corrosion Inhibition Efficiencies of Polymethacrylic Acid and Substituted Polymethacrylic Acid on Aluminium in 0.3M NaOH, ***ECS J. Solid State Technol.,*** 10, 101004, **2021**.

43. Madhuvathani Saminathan, Jothilal Palraj, Prince Wesley, Manojkumar Moorthy, Ravikirana, and Suresh Perumal*, Thermoelectric properties of p-type Si-rich Higher Manganese Silicides for mid-temperature applications, ***Mater. Lett.,*** 302,130444, **2021**.
44. John. D. Rodney, S. Deepapriya, M. Cyril Robinson, C. Justin Raj, Suresh Perumal, Byung C.K, S. Krishnan, S. Jerome Das, Dysprosium doped copper oxide ($Cu_{1-x}Dy_xO$) nanoparticles enabled bifunctional electrode for overall water splitting, ***Int. J. Hydrog. Energy,*** 46, 27585-27596, **2021**.
45. Lourdu Jame, Joselin Retna Kumar, Suresh Perumal, Jeyashree, Manojkumar Moorthy, Experimental Investigation of Thermoelectric power generator using D-Mannitol Phase change material for transient heat recovery. ***ECS J. Solid State. Technol.,*** 10, 061005, **2021**.
46. J. D. Rodney, S. Deepapriya, M. Cyril Robinson, S. Jerome Das, Suresh Perumal, Byung Chul Kim, C. Justin Raj, $Cu_{1-x}RE_xO$ (RE=La,Dy) decorated dendritic CuS nanoarrays for highly efficient splitting of seawater into hydrogen and oxygen fuels, ***Appl. Mater. Today,*** 24, 101079, **2021**.
47. Vikrant Trivedi, Manjusha Battabyal, Suresh Perumal, Avnee Chauhan, Dillip K. Satapathy, Budaraju Srinivasa Murty, and Raghavan Gopalan, Effect of Refractory Tantalum Metal Filling on the Microstructure and Thermoelectric Properties of Co_4Sb_{12} Skutterudites, ***ACS Omega,*** 6 (5), 3900-3909, **2021**.
48. Manivannan Kokarneswaran, Prakash Selvaraj, Thennarasan Ashokan, Suresh Perumal, Pathikumar Sellappan, Kandhasamy Durai Murugan, Sivanantham Ramalingam, Nagaboopathy Mohan, and Vijayanand Chandrasekaran, Discovery of Carbon nanotubes in 6th Century B.C. potteries from Keeladi, India. ***Sci. Rep.,*** 10(1), 1-6, ***2020***.
49. John D Rodney, S. Deepapriya, M. Cyril Robinson; C. Justin Raj; Suresh Perumal; Byung Chul Kim, Lanthanum doped copper oxide nanoparticles enabled proficient bi-functional electrocatalyst for overall water splitting, ***Int. J. Hydrog. Energy,*** 45 (46), 24684-24696, ***2020***.
50. Mathan Kumar, P. Bharathi, P. Palanisamy, S. Harish, Suresh Perumal, Ganeshkumar Mani, P. Dhivya, S. Ponnusamy, C. Muthamizhchelvan and M. Krishnamohan, Synthesis and Functional Properties of G-Doped WO3/TiO2 composited for sensing applications, **Mater. Sci. Semicon. Proc**., 105, 104732, **2020**.
51. R. S. Nathiya, Suresh Perumal, Malathy Moorthy, Vajjiravel Murugesan, Rajavel Rangappan, V. Raj, "Synthesis, characterization and inhibition performance of Schiff bases for aluminium corrosion in $1MH_2SO_4$ solution", ***J. Bio Tribo Corros***., 3, 6, ***2019***.
52. S. Perumal*, S. Gorsse, U. Ail, M. Prakasam, P. Rajasekar, A. M. Umarji, Enhanced Thermoelectric Figure of Merit in Nano-structured Si dispersed Higher Manganese Silicide", ***Mater. Sci. in Semicon. Proc.,*** 104, 104649, **2019**.
53. R. S. Nathiya, Suresh Perumal, Vajjiravel Murugesan, V. Raj, Evaluation of extracts of Borassus flabellifer dust as green inhibitors for aluminium corrosion in acidic media, ***Mater. Sci. in Semicon***. Proc., 104, 104674, ***2019***.
54. Suresh Perumal, Manisha Samanta, Tanmoy Ghosh, U. Sandhya Shenoy, Anil K. Bohra, Shovit Bhattacharya. Ajay Singh, Umesh V. Waghmare, and Kanishka Biswas, Realization of High Thermoelectric Figure of Merit in GeTe by Complementary Co-doping of Bi and In, ***Joule***, 3, 2565-2580, ***2019.***
55. S. Roychowdhury, M. Samanta, S. Perumal and K. Biswas, Germanium Chalcogenide Thermoelectrics: Electronic Structure Modulation and Low Lattice Thermal Conductivity, ***Chem. Mater.,*** 30, 5799-5813, ***2018.***
56. R. S. Nathiya, S. Perumal, V. Murugesan, V. Raj, Expired drugs: Environmentally safe inhibitors for aluminium corrosion in 1M H_2SO_4. ***J. Bio Tribo Corros.,*** 4 (1), 4, **2018.**
57. S. Perumal, P.Bellare, U. S. Sandhya, U. Waghmare, K. Biswas, Low Thermal Conductivity and High Thermoelectric performance in Sb and Bi co-doped GeTe: Complemetary Effect of Band Convergence and Nanostructuring. ***Chem. Mater.,*** 29, 10426-10435, ***2017.***
58. R. S. Nathiya, S. Perumal, V. Murugesan, P. M. Anbarasan, V. Raj, Agarose as an Efficient Inhibitor for Aluminium Corrosion in Acidic Medium: An Experimental and Theoretical Study. ***J. Bio Tribo Corros.,*** 3, 44, **2017.**
59. R. Suresh, V. Ponnuswamy, C. Sankar, M. Manickam, S. Venkatesan, S. Perumal, NiO nanoflakes: Effect of anions on the structural, optical, morphological and magnetic properties, ***J. Magn. Magn. Mater.,*** 441, 787-794, ***2017.***
60. M. Samanta, S. Roychowdhury, J.Ghatak, S. Perumal and K. Biswas, Ultrahigh Average Thermoelectric Figure of Merit, Low Lattice Thermal Conductivity and Enhanced Micro-hardness in Nanostructured $(GeTe)_x(AgSbSe_2)_{100-x}$, ***Chem. Eur. J.,*** 23, 1-7, **2017.**
61. S. Roychowdhury, R. Panigrahi, S. Perumal, and K. Biswas, Ultrahigh Thermoelectric Figure of Merit and Enhanced Mechanical Stability of p-type $AgSb_{1-x}Zn_xTe_2$, ***ACS Energy Lett.,*** 2, 349-356, ***2017.***

62. A. Banik, B. Vishal, S. Perumal, R. Datta, and K. Biswas; The origin of low thermal conductivity in $Sn_{1-x}Sb_xTe$: phonon scattering via layered intergrowth nanostructures, ***Energy Environ. Sci.,*** 9, 2011-2019, ***2016.***
63. S. Perumal, S. R. Choudury, K. Biswas, Reduction of thermal conductivity through nanostructuring enhances the thermoelectric figure of merit in $Ge_{1-x}Bi_xTe$, ***Inorg. Chem. Front.,*** 3, 125-132, ***2016.***
64. S. Perumal, S. R. Choudury, K. Biswas, High Performance thermoelectric materials and devices based on GeTe, ***J. Mater. Chem. C.,*** 4, 7520-7536, ***2016.***
65. S. Perumal, S. R. Choudury, D. S. Negi, R. Datta, K. Biswas, High Thermoelectric performance and Enhanced Mechanical Stability in Pb-free p-type $Ge_{1-x}Sb_xTe$, ***Chem. Mater.,*** 27, 7171-7178, ***2015.***
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Declaration

I hereby declare that the information and documents which I have mentioned above are true to the best of my knowledge and belief.

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